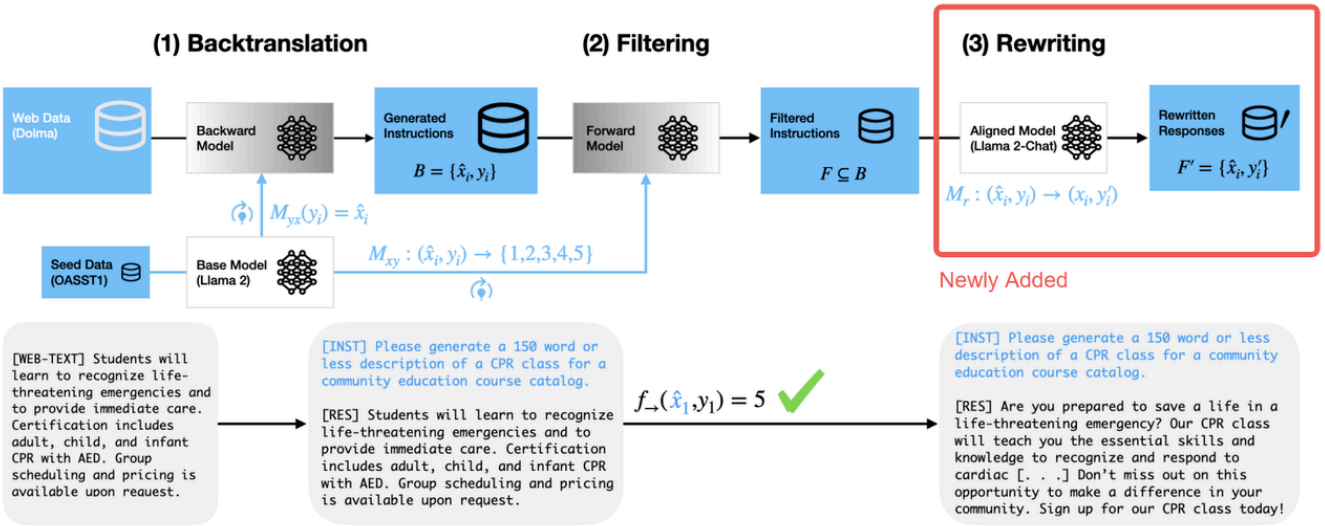
Better Alignment with Instruction Back-and-Forth Translation

> #Dataset #Finetuning

> *This framework is based on the previous [work](#).*



- What problem does this framework solve?
  - Similar to the previous work.
  - In its previous work, a close-source dataset, `ClueWeb`, was used, which needs to be paid.
  - Plus, `ClueWeb`, along with other datasets from the internet, contains some useless information, such as HTML structures and user-oriented writing style. This will degrade the quality of the response generated by the model.
- What does this framework do?
  - Instead of using `ClueWeb`, the new framework used an open dataset `Dolma`.
  - Add a new step after the data filtering process: rewrite. A finetuned LLM will be used to re-write the answer y based on generated $\hat{x}$ to form y', during which useless tokens and structural marks will be removed, making it more closely resemble the response style of an AI assistant and eliminating the adverse effect from the HTML structures.
    - Originally, a manual coded HTML cleaner function was applied to the data. However, this will heavily rely on the robustness of the cleaning logic. Using a pre-trained LLM can solve this problem once and for all.
  - Then, continue the processes mentioned in the previous work given the dataset $F^* = \{\hat{x}_i, y_i'\}$.
  - It is found that cleaning > filtering. Even if we remove the filtering process and rewrite ALL generated $(\hat{x}, y)$ pairs, the performance is still better than filtering only framework.
- Experiment Results
  - Superior Performance: When trained on the same dataset volume, the Llama-2 model fine-tuned using this method outperformed the baseline method relying solely on filtered network data on

AlpacaEval. It also surpassed current mainstream synthetic instruction datasets such as OpenOrca, ShareGPT, Evol-Instruct, and Alpaca-GPT4.
    - Criticality of Rewriting: Ablation experiments demonstrate that “response rewriting” (Step 3) plays a more significant role in enhancing data quality than “data filtering” (Step 2). Even when rewriting unfiltered initial text directly yields better results than filtering alone without rewriting.
    - Balancing Diversity and Quality: MAUVE distribution analysis reveals that rewritten responses occupy an intermediate position in the embedding space between “original web text” and purely “model distillation outputs.” This indicates the method successfully preserves the diversity and complexity of web data while ensuring high quality for direct alignment tasks through model rewriting.
- My Thought
    - *The same as the previous work.*
- Work Cited
    - Paper
    - NotebookLM
    - Chat History